

Raúl Díaz, PhD in Biological Sciences, Eng.

Córdoba, Argentina | +54 9 351 6474737 | raulenriquedd@hotmail.com

[GitHub](#)

[ResearchGate](#)

ORCID: 0000-0002-2674-1531

PROFESSIONAL SUMMARY

Data Scientist and Computational Biologist with a PhD in Biological Sciences and over 7 years of experience in data-intensive research. Proficient in R and Python for advanced statistical analysis, predictive modeling, and complex data visualization in the life sciences and environmental domains. My background bridges rigorous scientific methodology with the ability to build robust, automated analysis pipelines. I am seeking to apply my analytical expertise in an agile technology development environment, contributing to AI and data science projects for clients in the scientific and biotech sectors.

TECHNICAL SKILLS

- **Programming Languages:** R (Expert): tidyverse, Bioconductor, linear models, Shiny (basic). Python (Proficient): Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.
- **Data Analysis & Statistics:** Statistical Modeling (GLM, Mixed Models), Multivariate Statistics (PCA, NMDS, PERMANOVA, RDA), Experimental Design & Hypothesis Testing, Bayesian Methods (Introductory).
- **Scientific Domain (Bioinformatics):** Chemometrics (HPLC Data Modeling), Geographic Information Systems (GIS) & Spatial Data, Ecological Niche Modeling (Maxent, introductory).

PROFESSIONAL EXPERIENCE

Doctoral Fellow

Center for Ecology and Renewable Natural Resources (CERNAR) / CONICET | Córdoba, Argentina

2018 – 2023

- **Predictive Modeling in Mountain Ecosystems:** Designed and executed a research project integrating high-dimensional field data. Developed predictive models (Multiple regressions) in R and Python to quantify climate change impacts on lichen functional traits, resulting in 3 publications in international peer-reviewed journals (Q1/Q2).
- **Analysis Pipelines & Automation:** Built automated workflows in R and Bash for processing raw ecological and physiological data, reducing analysis time by 40% and ensuring full experimental reproducibility.

- Community Statistical Analysis: Applied multivariate statistical techniques (PCA, NMDS, PERMANOVA) in R to decompose biodiversity patterns along elevational gradients, transforming complex ecological hypotheses into quantifiable results and high-impact visualizations for academic stakeholders.
- Methodological Tool Development: Co-led the creation of a low-cost method for measuring thallus hydrophobicity, coding the resulting image analysis pipeline in R for feature extraction, yielding a published methodological tool (Applications in Plant Sciences, 2023).

Environmental Data Consultant

INICO1-19 Project, Faculty of Exact Sciences, National University of Asunción | Paraguay
2024 – Present

- Consulting engagement focused on structuring and analyzing biodiversity data. Implement R scripts for cleaning, validation, and Exploratory Data Analysis (EDA) of large biological record datasets, generating automated reports to support the research team's decision-making process.

Geospatial Data Analyst (Undergraduate Intern)

Faculty of Exact Sciences, National University of Asunción | Paraguay
2016 – 2017

- Developed a geospatial and climatic data analysis project using Excel. Processed 30-year precipitation datasets (time series) and conducted trend analysis to model a watershed's hydrological behavior, documenting the results in a scientific publication.

EDUCATION

- Ph.D. in Biological Sciences | National University of Córdoba, Argentina (2023)
- B.Sc. in Environmental Engineering | National University of Asunción, Paraguay (2018)

ADDITIONAL TECHNICAL TRAINING (Selected)

- Programming with Python | REDES Center, Argentina (2024). 40 hrs.
- Advanced Statistical Methods in Ecology and Evolution | University of the Republic, Uruguay (2021, 120 hrs)
- Niche and Distribution Modeling with Maxent | National University of Córdoba, Argentina (2019, 40 hrs)
- Introduction to R: Linear Models and Programming Fundamentals | National University of Córdoba, Argentina (2019, 40 hrs)

- Community Ecology Analysis in R | National University of the Littoral, Argentina (2018, 40 hrs)

LANGUAGES

- Spanish: Native
- English: Proficient (Reading, Writing, Listening) – Intermediate (Speaking). (Equivalent to B2/C1 technical/scientific level).
- Guaraní: Intermediate